



DIGITAL TECHNOLOGY

Academic Year 2024-25

Prof. Barbara PERNICI

Exam 26/1/2026

Total time 1.30 h

Last name

First name

Matricola/Person code

Signature

Please remember that:

- *The exam is closed books.*
- *The use of cellular phones or any other electronic devices during the exam is forbidden.*

It is necessary to answer at least partially all the questions for a positive evaluation.

Given answers should be explained, lists of bullet items are insufficient to answer a question.

Please write the answer for each question on a separate piece of paper.

Question 1 [11 points]

Explain the main benefits related to Agile methodologies, focusing on how uncertainty is managed compared with the traditional project management methodology.

Question 2 [11 points]

Illustrate the localization techniques for objects.

[please turn over]

Question 3 [11 points]

Consider the following fragment of Python code:

```
import pandas as pd

data = {
    "name": ["Mr. White", "Mr. Blonde", "Mr. White", "Mr. Pink", "Mr. Blonde"],
    "hours": [5, 8, 3, 8, 6]
}

df = pd.DataFrame(data)

def compute_pay(h):
    if h >= 8:
        return h * 15
    else:
        return h * 10

df["pay"] = df["hours"].apply(compute_pay)
expenses = df.pay.sum()
```

Questions:

- 1) What is the content of the *expenses* variable after running all the code?
- 2) Modify the *compute_pay* function so that only the portion of hours strictly over 8 are paid at 15 per hour, while keeping the rest of the logic unchanged.
- 3) Compute the pay per person.